

Md Anisur Rahman Chowdhury

Researcher in Cloud Security, Serverless Systems, and Network Infrastructure Protection

Graduate Research Assistant developing first-author and collaborative research in cloud security, serverless systems, Zero Trust architecture, physical and cloud network infrastructure protection, and practical AI-assisted defense.

Winnetka, California, USA / Anywhere in USA | +1 814-737-5770 | engr.aanis@gmail.com | chowdhur014@gannon.edu

LinkedIn: [linkedin.com/in/md-anisur-rahman-chowdhury-15862420a](https://www.linkedin.com/in/md-anisur-rahman-chowdhury-15862420a) | GitHub: github.com/ANIS151993 | Google Scholar: scholar.google.com/citations?user=NQyywPoAAAAJ | ResearchGate: researchgate.net/profile/Md-Anisur-Rahman-Chowdhury | Portfolio: marcbd.com

RESEARCH PROFILE

Graduate Research Assistant in the Computer and Information Science Department at Gannon University working on cloud security, serverless systems, Zero Trust enforcement, and physical and cloud network infrastructure protection.

First-author work covers the full research cycle: problem framing, literature review, methodology design, data collection, prototype implementation, experimental evaluation, and manuscript preparation.

Published first-author and co-authored work spans serverless security, edge computing, AI-enabled business systems, IoT intrusion detection, adaptive network security, digital forensics, and applied intelligent systems.

Seven years of enterprise IT and ISP experience keep the research direction grounded in deployable systems, measurable performance, and operational security requirements.

RESEARCH INTERESTS

Physical and cloud network infrastructure security; Serverless security and intelligent firewall architecture; Zero Trust enforcement and continuous verification; Federated threat intelligence and privacy-preserving defense; Secure distributed systems and edge computing; AI-driven intrusion detection and infrastructure automation; and Reliable LLM systems and distributed agent networks.

EDUCATION

Master of Science in Information Technology

Gannon University, Erie, Pennsylvania, USA | Aug 2024 - May 9, 2026 (Expected)

Graduate Research Assistant in the Computer and Information Science Department. GPA: 3.95/4.00. Coursework includes cloud computing, advanced networking, cybersecurity, distributed systems, and analytics.

Bachelor of Science in Electronics and Telecommunication Engineering

University of Science and Technology Chittagong (USTC), Bangladesh | 2017

Built the engineering foundation for later work in communications, networking, systems, and infrastructure security.

RESEARCH AND ACADEMIC EXPERIENCE

IT Systems and Operations Engineer (Part-Time, OPT)

Adel Group Inc, IT Department, Winnetka, California, USA | Jul 2026 - Present

- Support enterprise IT systems and network infrastructure, manage endpoint protection, access control, and firewall policy enforcement, and pilot helpdesk automation workflows.
- Keep research grounded in real production systems by working directly with the operational security and infrastructure concerns that motivate it.

Graduate Research Assistant

Gannon University, Computer and Information Science Department, Erie, Pennsylvania, USA | Aug 2025 - Present

- Conduct research under Dr. Kefei Wang on cloud systems, AI-driven security, distributed infrastructure protection, and secure network design.
- Lead first-author studies from idea formation through methodology design, implementation, evaluation, and manuscript preparation.
- Build reproducible research prototypes around serverless firewalls, Zero Trust automation, threat intelligence integration, distributed AI, and dependable LLM workflows.
- Prepare technical writing, experimental summaries, and publication-ready material for conference and journal submission.

Cybersecurity Lecturer (CPT)

TAAS Services LLC, Remote, USA | Aug 2025 - Dec 2025

- Delivered practical instruction in network defense, firewall configuration, threat detection, and vulnerability management.
- Strengthened teaching, mentoring, and technical communication skills that support research presentation and academic collaboration.
- Co-authored research with Dr. Kefei Wang on AI-enabled enterprise systems, cloud-based CRM optimization, edge resource allocation, and serverless firewall architectures.
- Contribute literature review, experiment planning, implementation support, results analysis, and manuscript refinement in collaborative projects.

RESEARCH SYSTEMS AND REPRODUCIBLE PROJECTS

- **Serverless Intelligent Firewall Research Series.** A connected line of research systems exploring AI-driven firewall behavior, cross-cloud adaptation, REST API threat intelligence, and federated Zero Trust protection.
- **Distributed AI Multi-Agent Ensemble.** A local distributed LLM system for complex problem solving and collaborative reasoning across multiple agents.
- **Self-Correcting LLM Mathematical Reasoning.** A three-stage framework for improving LLM reliability through review, correction, and answer refinement.
- **LLM-Powered Automated Penetration Testing Platform (APTS).** A self-hosted LLM orchestration system for autonomous penetration testing with three-layer scope enforcement and tamper-evident cryptographic audit chains for authorized, auditable security testing.
- **DataMentor / Notebook Studio.** A reproducible notebook and CSV workflow environment that supports experimentation, analysis, and research reporting.

PUBLISHED FIRST-AUTHOR PAPERS

- Towards a Serverless Intelligent Firewall: AI-Driven Security, and Zero-Trust Architectures. 2025 IEEE 12th International Conference on Cyber Security and Cloud Computing (CSCloud).
- Auction-Based Dynamic Resource Allocation for Optimized Edge Computing in Distributed Networks. 2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS).
- AI and Cloud Computing in Business Systems: A Hybrid Model for Enhancing Enterprise Resource Planning. 2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS).
- Deepfake Detection in MIS: Leveraging DenseNet and Multi-Scale Information for Enhanced Digital Forensics. IEEE COMPAS 2025.
- Fed Threat-X: A Privacy-Preserving Federated Threat Intelligence Framework for Multi-Cloud Cybersecurity. IEEE QPAIN 2026.

PUBLISHED CO-AUTHORED PAPERS

- Cloud-Based CRM Systems Enhanced by AI and Graph Theory: A Hybrid Model for Optimizing Customer Engagement. IEEE ISAECT 2025.
- Enhancing Signature-Based Intrusive Detection System (IDS) for IoT Networks Using Machine Learning Algorithm. IEEE QPAIN 2025.
- AI-Enhanced Adaptive Network Security for 6G and Edge Computing. IEEE QPAIN 2025.
- Detecting misinformation with multimodal AI: leveraging vision and NLP for fact-checking. IEEE QPAIN 2025.
- Driving Industry 4.0 with Digital Twins: Enhancing Predictive Maintenance and Operational Performance Through IoT and Machine Learning. IEEE QPAIN 2025.
- Leveraging Machine Learning and NLP for Adaptive Education Systems: A Personalized Approach for Children. IEEE QPAIN 2025.
- Attention-Enhanced U-Net Models for Breast Cancer Image Analysis: A Comparative Study. IEEE ECCE 2025.
- Enhanced Brain Tumor Detection Using Finetuned Transfer Learning Models: Achieving Superior Accuracy with Xception. IEEE ECCE 2025.
- Fed-ZTA-Transformer: A Privacy-Preserving Federated Framework for Continuous Verification in Zero Trust Architectures. IEEE QPAIN 2026.

- A Collaborative Hybrid CNN-LSTM Framework for Real-Time Multichannel Retail Demand Forecasting Using Edge-Cloud Computing. IEEE QPAIN 2026.

ACCEPTED AND UNDER REVIEW

- Towards a Serverless Intelligent Firewall: Integrating Cross-Cloud Adaptation, AI-Driven Security, and Zero-Trust Architectures. IEEE SmartCloud 2026. Accepted, awaiting publication. First Author.
- Teaching Large Language Models to Think Twice: A Three-Stage Framework for Self-Correcting Mathematical Reasoning. American CSE AIR-RES / CAC 2026. Accepted, awaiting publication. First Author.
- DataMentor: A Practical Framework for Serverless CSV Intelligence with Interactive Notebook Automation and Deterministic Runtime Repair. IEEE ISAIA 2026 (Submission 61). Under review. First Author.

IN PREPARATION

- Autonomous Self-Learning Serverless Intelligent Firewall: Integrating REST API-Driven Open-Source Threat Intelligence, Multi-Paradigm Machine Learning, and Federated Zero-Trust Architectures. Target: Q1 Journal. First Author.
- Self-Hosted LLM Orchestration for Autonomous Penetration Testing with Three-Layer Scope Enforcement and Tamper-Evident Cryptographic Audit Chains. Target: IEEE Conference. First Author.
- A Hybrid AI and Machine Learning Approach for Real-Time Anomaly Detection in IoT Networks. Target: IEEE Conference. First Author.
- Building a Team of AI Models: A Literature Review on Distributed Agent Networks. Target: IEEE Conference. First Author.

TECHNICAL TOOLS FOR RESEARCH

- **Programming:** Python, Java, JavaScript, C++, SQL, PL/SQL, JSON
- **Research and Systems:** AWS, Linux, Windows Server, Git, LaTeX-oriented writing workflow, serverless architectures, virtualization, network monitoring
- **Security and Networking:** firewalls, VPN, Active Directory, Zero Trust concepts, threat intelligence pipelines, traffic analysis, distributed network design
- **Data and Evaluation:** data collection, preprocessing, model training, experimental analysis, reproducible notebooks, technical reporting

RESEARCH STRENGTHS

- First-author ownership across problem framing, literature review, methodology design, implementation, evaluation, and manuscript development.
- Research agenda grounded in cloud security, serverless systems, Zero Trust architecture, federated defense, and physical and cloud network infrastructure protection.
- Ability to translate seven years of enterprise IT and ISP experience into research questions with operational relevance and deployable outcomes.
- Comfort working in collaborative research settings that require reproducible systems, careful experimentation, and clear technical communication.

RESEARCH GOAL

Seeking a fully funded PhD position for Fall 2026 in cloud security, serverless systems, network infrastructure protection, or a closely related cybersecurity area. Long-term research goal is to produce rigorous work that advances academic knowledge while improving how real systems are designed, secured, and operated.